

Introduction

Every minute, a plane is taking off or landing, connecting countless destinations across the globe. The immense volume of air traffic generates vast amounts of data, offering valuable insights into various aspects of aviation. As part of our data science project, we set out to uncover intriguing patterns in U.S. air traffic. Our focus was on identifying variables that influence different parts of flight travel, including delays, the impact of the pandemic, airfare prices, and the busiest routes.

Method

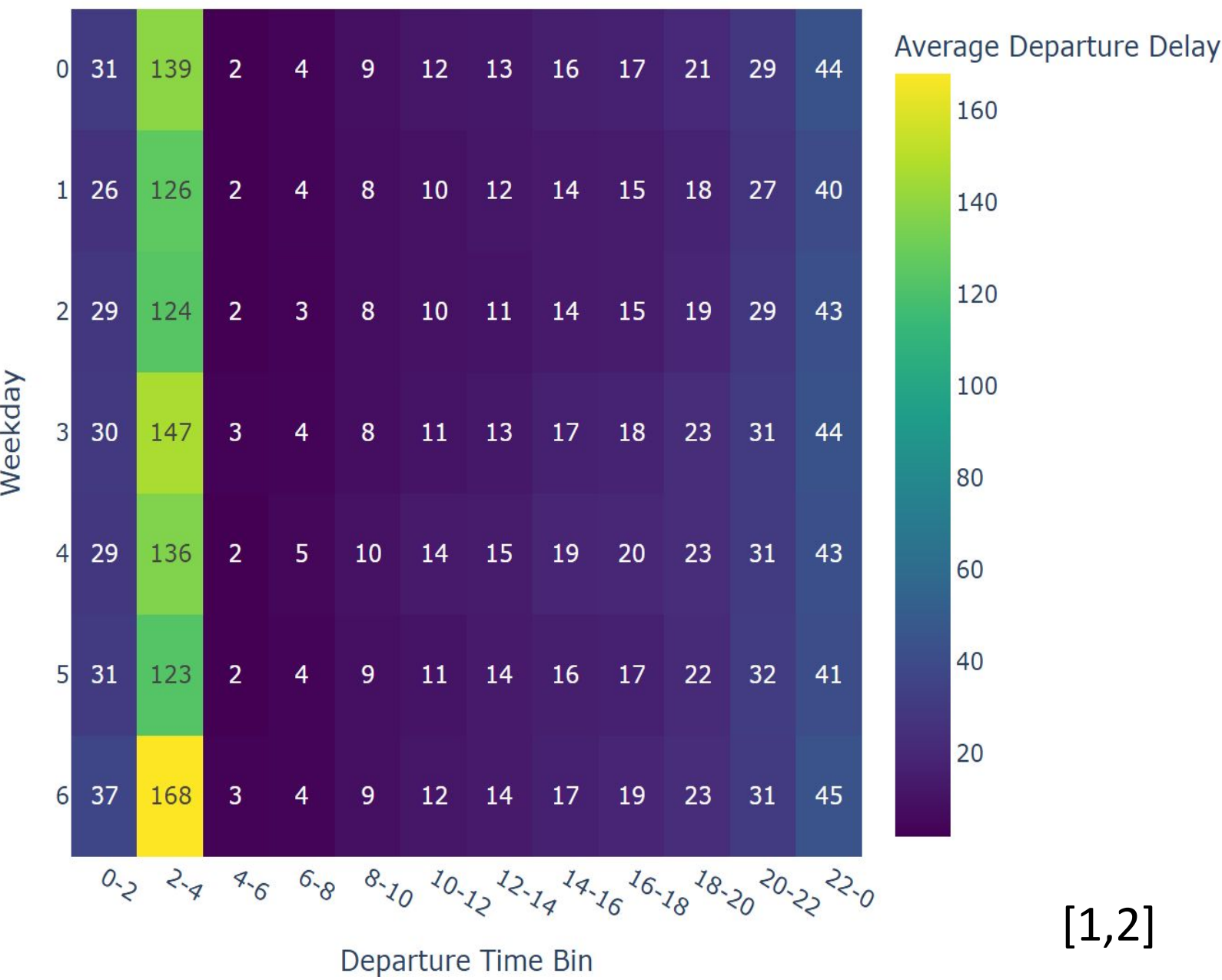
Our research methods involved collecting and processing various data sources to address several different research questions.

1. We filtered data containing international and domestic flight information, complemented by IATA codes and geolocations, and focused on the top 130 U.S. airports.
2. We integrated API data on COVID-19 alongside data on total departures to analyze the pandemic’s impact on air traffic.
3. We utilized monthly CPI data from the past 20 years to investigate economic influences. Additionally, we analyzed flight prices using three separate CSV files: one for airport-specific prices across different years, and two more focusing on price increases and decreases. Another CSV file was used to extract fuel prices.

Results

How do flight delays vary by weekday and time of day, and are there any patterns?

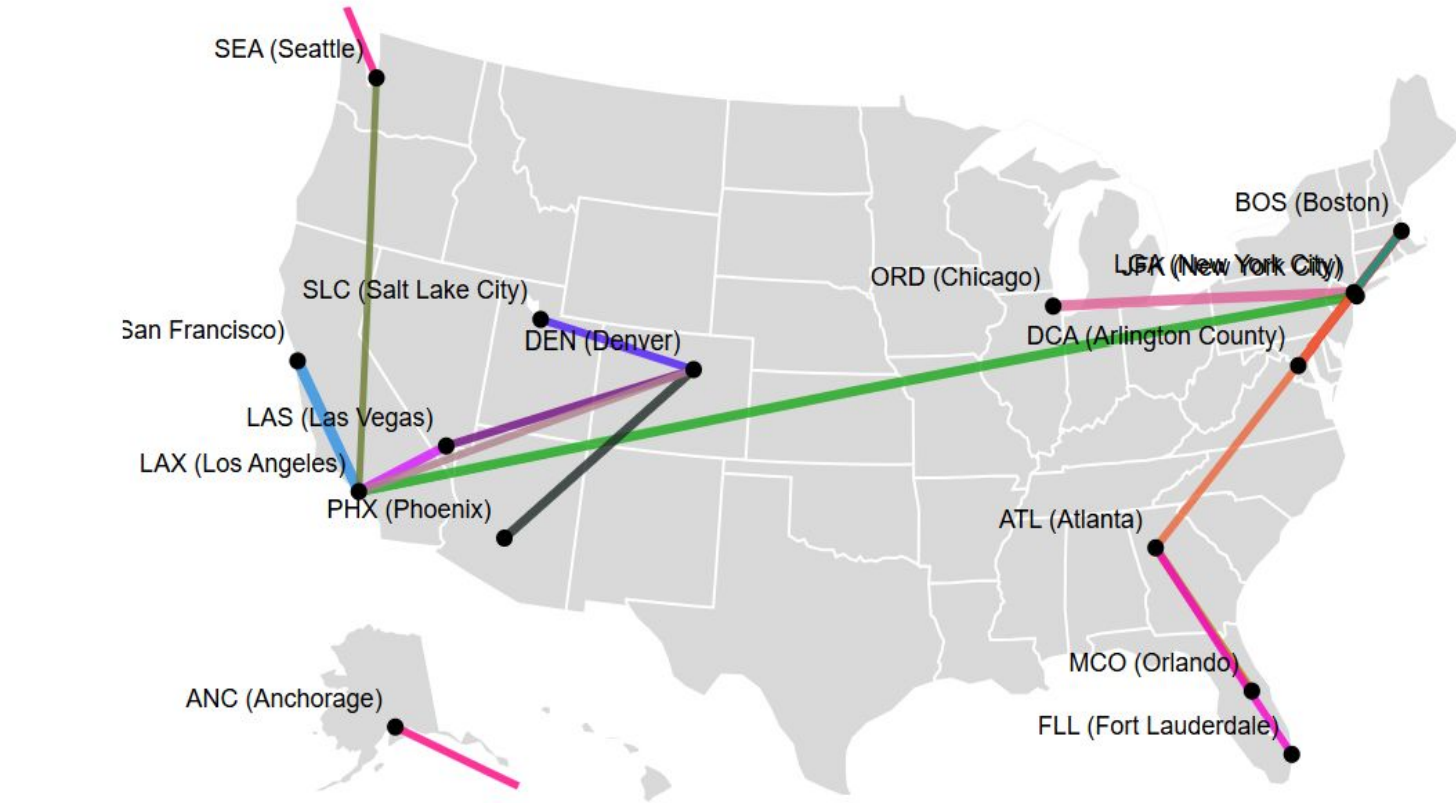
Heatmap of Average Departure Delay by Weekday and Departure Time Bin



[1,2]

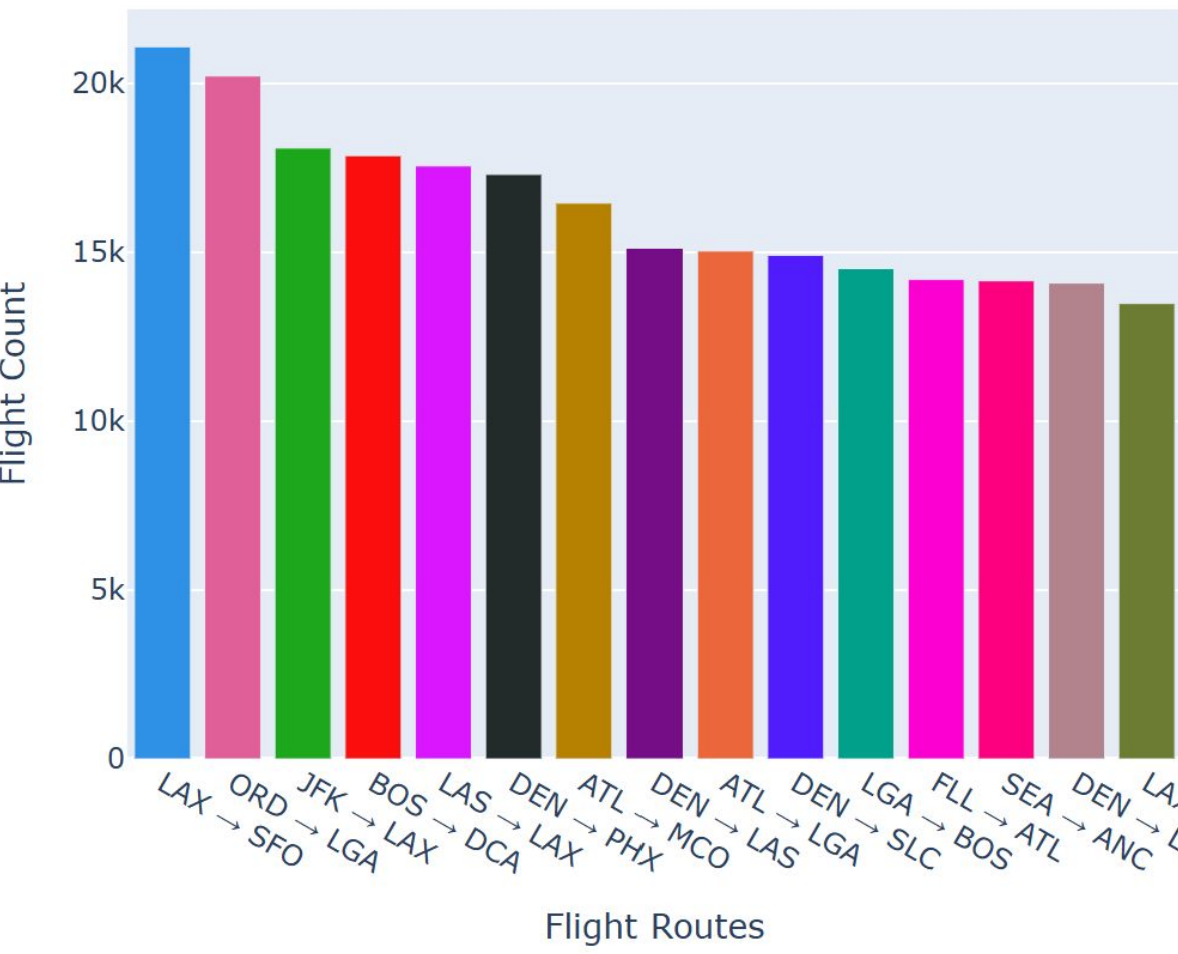
The highest delay for departures (without arrivals) occurs predominantly between 2 AM and 4 AM on Sunday, followed by Thursday during the same time period. The other weekdays also show similarly high delays. The most significant factor influencing delays appears to be the time rather than the specific day. The second highest delay occurs between 10 PM and 12 AM, but it is only about a quarter as severe compared to the former.

Which domestic flight routes have the highest traffic in USA?



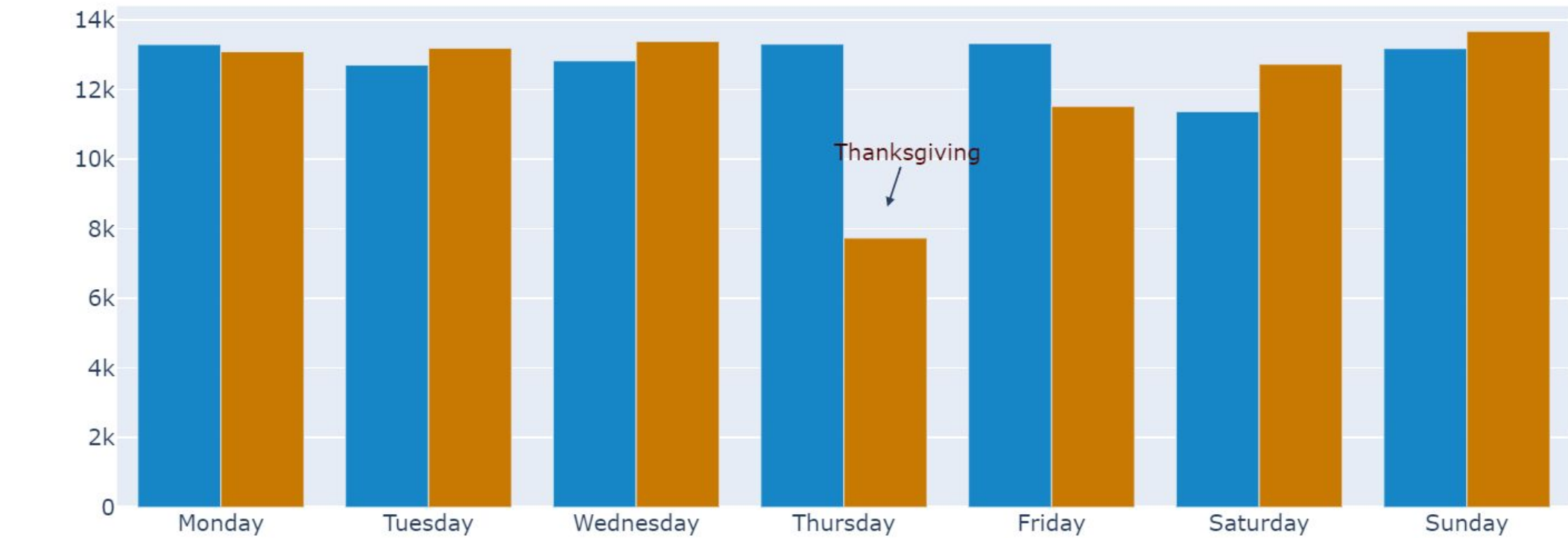
[1,2,3]

Flight Count for Top 15 Routes in 2024



[1,2,3]

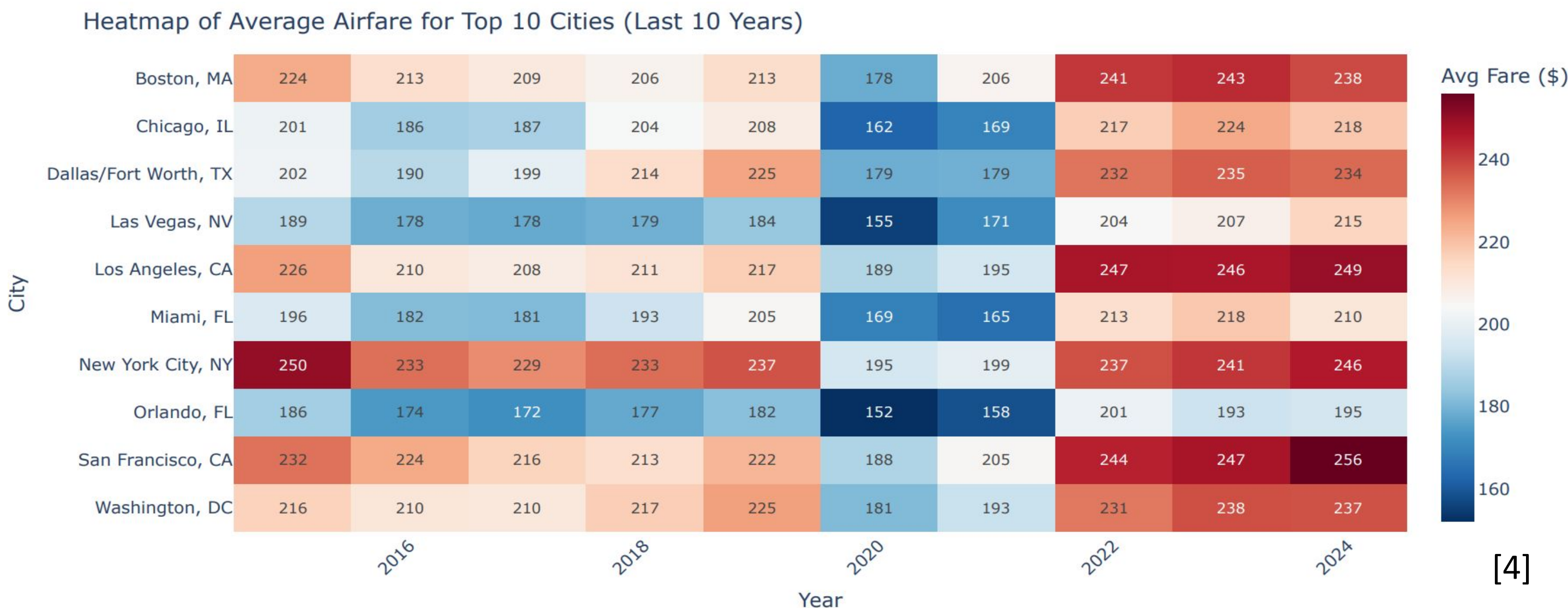
How did Thanksgiving affect the air traffic?



[1,2]

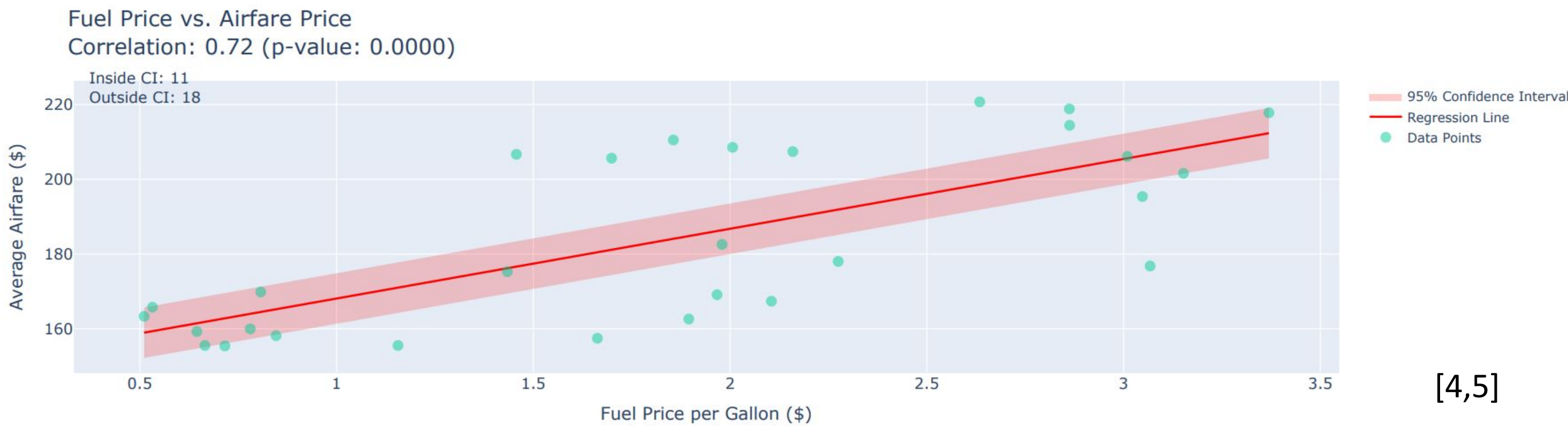
On Thanksgiving in both 2023 (see figure above) and 2024, domestic air traffic decreased by nearly 50%, contrary to our expectation of a significant increase in traffic 2-3 days before the holiday.

How have flight prices changed over time and what affects the price?



[4]

Airfare prices dropped sharply in 2020 and 2021 due to the impact of COVID-19, then began rising significantly from 2022 onward. The heatmap also reveals consistent price differences between cities, with New York and San Francisco generally having higher fares than more affordable destinations like Orlando and Las Vegas.

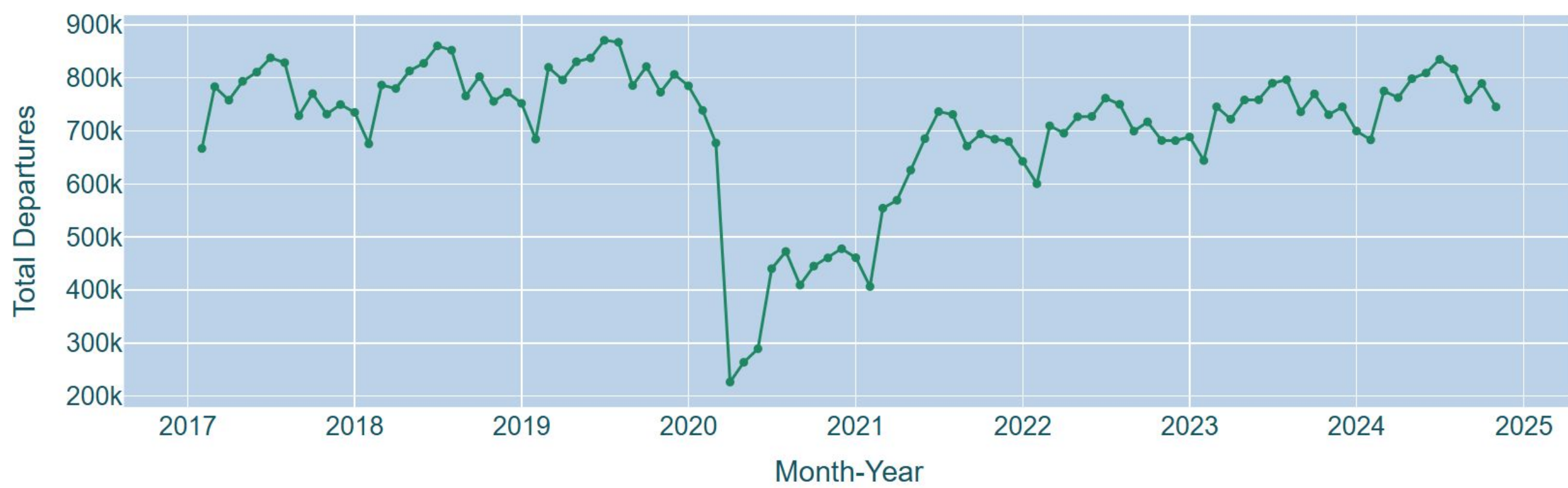


[4,5]

This scatterplot shows a strong link ($r = 0.72$) between fuel prices and airfare - when fuel costs rise, so do ticket prices. The trend line shows the overall pattern, but many points fall outside the 95% confidence range, hinting at other factors at play.

How did the number of passengers at major airports change pre, during and post COVID-19?

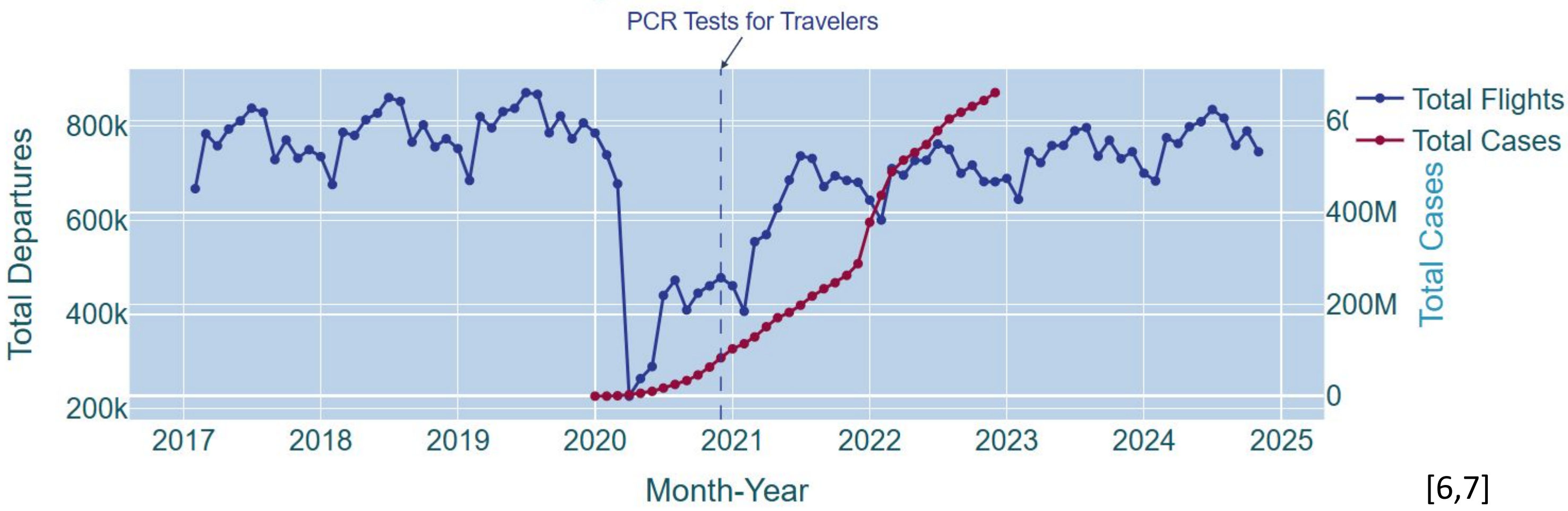
Total Departures



[6]

The graph represents total departures from 2017 to 2025. We can evidently see a deep dive starting from 2020, reducing the total flight numbers from ~800k to ~200k. Just as you thought, COVID-19 impacted the air travel.

Correlation Between Total Flights and COVID-19 Cases



[6,7]

This graph shows the correlation between the total number of flights in the United States (as depicted above) and global COVID-19 cases. As COVID-19 cases rose, flight numbers dropped drastically, which can be linked to the pandemic’s impact. The introduction of safety measures and regulations, such as PCR tests for travelers, helped maintain safer air travel despite rising case numbers. Over time, these measures supported the recovery of the air travel industry, as reflected in the gradual increase in flight numbers.

Bibliography

- [1] Liste der Verkehrsflughäfen in den Vereinigten Staaten – Wikipedia
- [2] BTS On-Time Performance
- [3] Airports/airports.csv at main · lxnrbzlz/Airports

- [4] Consumer Airfare Report: Table 2 - Top 1,000 City-Pair Markets | Department of Transportation
- [5] https://www.transtats.bts.gov/Fields.asp?gnoyr_VQ=FMH
- [6] OST_R | BTS | Transtats Data Tables
- [7] <https://covid-api.com>

